1 Homogeneous linear ODEs with constant coefficients

McMaster University

Mathematics and Statistics

$$\int_M d\omega = \int_{\partial M} \omega$$

Mathematics 2C03
Introduction to Differential Equations

Instructor: David Earn

Lecture 17
Homogeneous linear ODEs with constant coefficients
Thursday 22 October 2026

Announcements

- The TA is Illya Kierkosz (`kierkosi@mcmaster.ca`)
  - Office hours by appointment: e-mail to arrange a meeting.
- Course web site: `https://davidearn.github.io/math2c/`
- The course outline is on the Simple Syllabus web site.
- Avenue to Learn provides secure access to Echo360 live streams and lecture recordings.
- Assignments and class polls are on `childsmath`.
- **e-mail:** PLEASE always include 2C03 in the subject line of any message about this course, and PLEASE indicate what the issue is. For example:
  - `Subject: Math 2C03: confusion about assignment 1`
  - `Subject: Math 2C03: conflict with test window`
- Office hour: Thursdays, 9:30–10:20 am, after class.

Linear ODEs with Constant Coefficients

Trench §5.2: second-order equations.

Trench §9.2: higher-order equations.

Homogeneous, **1st** order, constant coefficients

$$L[y] = ay' + by = 0\,, \qquad a, b \in \mathbb{R} \tag{1}$$

If $a = 0$ then this is not a differential equation. So assume $a \neq 0$. Then Equation (1) is equivalent to

$$y' = -\frac{b}{a}y\,,$$

which has solution $y_c(x) = c\,e^{-(b/a)x}$ for any $c \in \mathbb{R}$.

Can knowing the solution to this 1st order linear ODE help us find solutions of higher order ODEs?

Stroke of genius: For higher order, linear, homogeneous ODEs, try a solution of the form e^{rx} and see if we can find a value of r that does yield a solution.

Homogeneous, **2nd** order, constant coefficients

$$L[y] \;=\; ay'' + by' + cy \;=\; 0\,, \qquad a, b, c \in \mathbb{R} \tag{2}$$

If $a = 0$ then we have the 1st order case. So assume $a \neq 0$.

Try solution of the form $y = e^{rx}$, *i.e.*, plug this guess into Equation (2). To that end:

$$\begin{aligned} y &= e^{rx} \\ \Longrightarrow\; y' &= \frac{\mathrm{d}}{\mathrm{d}x}(e^{rx}) = r\,e^{rx} \\ \Longrightarrow\; y'' &= \frac{\mathrm{d}}{\mathrm{d}x}(r\,e^{rx}) = r^2\,e^{rx} \end{aligned}$$

Homogeneous, **2nd** order, constant coefficients

$$L[y] = ay'' + by' + cy = 0, \qquad a, b, c \in \mathbb{R} \tag{2}$$

Thus, plugging $y = e^{rx}$ into Equation (2) we get

$$\begin{aligned} L[y] = L[e^{rx}] &= ar^2 e^{rx} + bre^{rx} + ce^{rx} = 0 \\ &\Longrightarrow e^{rx}(ar^2 + br + c) = 0 \\ &\Longrightarrow ar^2 + br + c = 0 \qquad \textbf{\textit{characteristic equation}} \\ &\Longrightarrow r = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \qquad (OK \ \because \ a \neq 0) \end{aligned}$$

What is the meaning of this solution for r?

Several cases to consider, depending on the sign of the ***discriminant*** $(b^2 - 4ac)$ ...

Homogeneous, **2nd** order, constant coefficients

$$L[y] = ay'' + by' + cy = 0, \qquad a, b, c \in \mathbb{R},\ a \neq 0. \tag{2}$$

The solutions of the characteristic equation ($ar^2 + br + c = 0$) are

$$r_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$$
$$r_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

so we appear to have two solutions to $L[y] = 0$, namely

$$y_1(x) = e^{r_1 x} \quad \text{and} \quad y_2(x) = e^{r_2 x}$$

Please do the poll *Distinct exponential solutions* on childsmath ...

Poll

- Go to
  https://www.childsmath.ca/childsa/forms/main_login.php
- Click on Math 2C03
- Click on Take Class Poll
- Fill in poll **Distinct exponential solutions**
- Submit.

Distinct exponential solutions

$$L[y] = ay'' + by' + cy = 0, \qquad a, b, c \in \mathbb{R}, \quad a \neq 0.$$

The solutions of the characteristic equation $(ar^2 + br + c = 0)$ are

$$r_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \qquad r_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a},$$

so we appear to have two solutions to $L[y] = 0$, namely

$$y_1(x) = e^{r_1 x} \qquad \text{and} \qquad y_2(x) = e^{r_2 x}.$$

A y_1 and y_2 are always distinct.

B y_1 and y_2 are never distinct.

C y_1 and y_2 are distinct if and only if $b^2 - 4ac \neq 0$.

D y_1 and y_2 are useful if and only if $b^2 - 4ac > 0$.

E y_1 and y_2 are meaningless if $b^2 - 4ac < 0$.

Answer: C is the only correct answer.

Homogeneous, **2nd** order, constant coefficients

$$L[y] = ay'' + by' + cy = 0, \qquad a, b, c \in \mathbb{R},\ a \neq 0. \tag{2}$$

- *characteristic equation*: $ar^2 + br + c = 0$
- *solutions*: $r_1 = \frac{-b+\sqrt{b^2-4ac}}{2a}$ $r_2 = \frac{-b-\sqrt{b^2-4ac}}{2a}$

What are the solutions to $L[y] = 0$? 3 cases...

Case 1: ***Distinct Real Roots***

$$b^2 - 4ac > 0 \implies r_1 \neq r_2, \quad r_1, r_2 \in \mathbb{R}$$

Case 2: ***Repeated Real Roots***

$$b^2 - 4ac = 0 \implies r_1 = r_2 = -\frac{b}{2a} \in \mathbb{R}$$

Case 3: ***Conjugate Complex Roots***

$$b^2 - 4ac < 0 \implies r_1 = \overline{r_2} \in \mathbb{C}$$

Why isn't there another case with $r_1 = r_2 \in \mathbb{C}$?

Answer: $r_1 = r_2 \implies b^2 - 4ac = 0.$

Homogeneous, **2nd** order, constant coefficients

$$L[y] = ay'' + by' + cy = 0, \qquad a, b, c \in \mathbb{R},\ a \neq 0. \tag{2}$$

- *characteristic equation*: $ar^2 + br + c = 0$
- *solutions*: $r_1 = \frac{-b+\sqrt{b^2-4ac}}{2a}$ $r_2 = \frac{-b-\sqrt{b^2-4ac}}{2a}$

Case 1: ***Distinct Real Roots*** $b^2 - 4ac > 0$, $r_1 \neq r_2$, $r_1, r_2 \in \mathbb{R}$

Solutions: $y_1 = e^{r_1 x}$ and $y_2 = e^{r_2 x}$

Are they linearly independent?

Wronskian: $$W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^{r_1 x} & e^{r_2 x} \\ r_1 e^{r_1 x} & r_2 e^{r_2 x} \end{vmatrix}$$

$$= (r_2 - r_1)e^{(r_1+r_2)x} \neq 0 \quad \text{for any } x \in \mathbb{R}$$

$\Longrightarrow$ y_1, y_2 linearly independent on $(-\infty, \infty)$

$\Longrightarrow$ $\{y_1, y_2\}$ is a fundamental solution set on $(-\infty, \infty)$

General solution: $y = c_1 e^{r_1 x} + c_2 e^{r_2 x}$, $c_1, c_2 \in \mathbb{R}$

Homogeneous, **2nd** order, constant coefficients

$$L[y] = ay'' + by' + cy = 0, \qquad a, b, c \in \mathbb{R},\ a \neq 0. \tag{2}$$

- *characteristic equation*: $ar^2 + br + c = 0$
- *solutions*: $r_1 = \frac{-b+\sqrt{b^2-4ac}}{2a}$ $r_2 = \frac{-b-\sqrt{b^2-4ac}}{2a}$

Case 2: ***Repeated Real Roots*** $b^2 - 4ac = 0$, $r_1 = r_2 \equiv r \in \mathbb{R}$

Solutions: $y_1 = e^{rx}$ and $y_2 = xe^{rx}$ (*Check!*)
Are they linearly independent?

Wronskian: $$W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^{rx} & xe^{rx} \\ re^{rx} & (rx+1)e^{rx} \end{vmatrix}$$
$$= (rx+1)e^{2rx} - (rx)e^{2rx} = e^{2rx} \neq 0 \quad \forall\, x \in \mathbb{R}$$

$\Longrightarrow$ y_1, y_2 linearly independent on $(-\infty, \infty)$

$\Longrightarrow$ $\{y_1, y_2\}$ is a fundamental solution set on $(-\infty, \infty)$

General solution: $y = (c_1 + c_2 x)e^{rx}$, $c_1, c_2 \in \mathbb{R}$

Homogeneous, **2nd** order, constant coefficients

$$L[y] = ay'' + by' + cy = 0, \qquad a, b, c \in \mathbb{R},\ a \neq 0. \tag{2}$$

- *characteristic equation*: $ar^2 + br + c = 0$
- *solutions*: $r_1 = \frac{-b+\sqrt{b^2-4ac}}{2a}$ $r_2 = \frac{-b-\sqrt{b^2-4ac}}{2a}$

Case 3: ***Conjugate Complex Roots*** $b^2 - 4ac < 0, \quad r_1 = \overline{r_2} \in \mathbb{C}$

Note: $b^2 - 4ac < 0 \implies \sqrt{b^2-4ac} = \sqrt{-|b^2-4ac|}$

$$= \sqrt{-1}\sqrt{|b^2-4ac|} = i\sqrt{|b^2-4ac|}$$

Therefore, we can write

$$r_1 = \lambda + i\omega, \quad r_2 = \lambda - i\omega$$

where $\lambda = -\frac{b}{2a} \in \mathbb{R}$ and $\omega = \frac{\sqrt{|b^2-4ac|}}{2a} \in \mathbb{R}$

Homogeneous, **2nd** order, constant coefficients

$$L[y] = ay'' + by' + cy = 0, \qquad a, b, c \in \mathbb{R},\ a \neq 0. \tag{2}$$

- *characteristic equation*: $ar^2 + br + c = 0$
- *solutions*: $r_1 = \frac{-b+\sqrt{b^2-4ac}}{2a}$ $r_2 = \frac{-b-\sqrt{b^2-4ac}}{2a}$

Case 3: ***Conjugate Complex Roots*** $b^2 - 4ac < 0$, $r_1 = \overline{r_2} \in \mathbb{C}$

Solutions: $e^{r_1 x} = e^{(\lambda+i\omega)x} = e^{\lambda x}e^{i\omega x}$ and
$e^{r_2 x} = e^{(\lambda-i\omega)x} = e^{\lambda x}e^{-i\omega x}$

But $e^{i\omega x}$ isn't a real-valued function (!).

What does this mean?

What should we do?

Clearly, we should do the poll
Real solutions from complex roots on childsmath

Poll

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  https://www.childsmath.ca/childsa/forms/main_login.php
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- Fill in poll **Real solutions from complex roots**
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Homogeneous, **2nd** order, constant coefficients

$$L[y] = ay'' + by' + cy = 0, \qquad a, b, c \in \mathbb{R},\ a \neq 0. \tag{2}$$

- *characteristic equation*: $ar^2 + br + c = 0$
- *solutions*: $r_1 = \frac{-b+\sqrt{b^2-4ac}}{2a}$ $r_2 = \frac{-b-\sqrt{b^2-4ac}}{2a}$

Case 3: ***Conjugate Complex Roots*** $b^2 - 4ac < 0$, $r_1 = \overline{r_2} \in \mathbb{C}$

Recall ***Euler's formula***: $e^{i\theta} = \cos\theta + i\sin\theta$

$$\therefore \quad e^{r_1 x} = e^{\lambda x}\left(\cos(\omega x) + i\sin(\omega x)\right)$$
$$e^{r_2 x} = e^{\lambda x}\left(\cos(\omega x) - i\sin(\omega x)\right)$$

Recall L is a linear operator, so linear combinations of solutions to $L[y] = 0$ are also solutions. Therefore, let

$$y_1 = \frac{1}{2}\left(e^{r_1 x} + e^{r_2 x}\right) = e^{\lambda x}\cos(\omega x),$$
$$y_2 = \frac{1}{2i}\left(e^{r_1 x} - e^{r_2 x}\right) = e^{\lambda x}\sin(\omega x)$$

Homogeneous, **2nd** order, constant coefficients

$$L[y] = ay'' + by' + cy = 0, \qquad a, b, c \in \mathbb{R},\ a \neq 0. \tag{2}$$

- *characteristic equation*: $ar^2 + br + c = 0$
- *solutions*: $r_1 = \frac{-b+\sqrt{b^2-4ac}}{2a}$ $r_2 = \frac{-b-\sqrt{b^2-4ac}}{2a}$

Case 3: ***Conjugate Complex Roots*** $b^2 - 4ac < 0$, $r_1 = \overline{r_2} \in \mathbb{C}$

Wronskian: $W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$

$$= \begin{vmatrix} e^{\lambda x}\cos(\omega x) & e^{\lambda x}\sin(\omega x) \\ e^{\lambda x}\big(\lambda\cos(\omega x) - \omega\sin(\omega x)\big) & e^{\lambda x}\big(\omega\cos(\omega x) + \lambda\sin(\omega x)\big) \end{vmatrix}$$

$\neq 0$ for $x = 0$ $\implies$ $\neq 0$ for all $x \in \mathbb{R}$

$\implies$ y_1, y_2 linearly independent on $(-\infty, \infty)$

$\implies$ $\{y_1, y_2\}$ is a fundamental solution set on $(-\infty, \infty)$

General solution: $y = e^{\lambda x}\big(c_1\cos(\omega x) + c_2\sin(\omega x)\big), \quad c_1, c_2 \in \mathbb{R}$

Homogeneous, **2nd** order, constant coefficients

Example ($y'' + 3y' + 2y = 0$)

Characteristic equation:

$$r^2 + 3r + 2 = 0 \implies (r+1)(r+2) = 0$$
$$\implies r_1 = -1,\ r_2 = -2 \qquad \textit{real, distinct}$$

Linearly independent solutions:

$$y_1 = e^{-x}, \qquad y_2 = e^{-2x}$$

General solution:

$$y = c_1 e^{-x} + c_2 e^{-2x}$$

Homogeneous, **2nd** order, constant coefficients

Example ($y'' + k^2 y = 0, \quad k > 0$)

Characteristic equation:

$$r^2 + k^2 = 0 \implies (r^2 - i^2 k^2) = 0 \implies (r^2 - (ik)^2) = 0$$
$$\implies (r + ik)(r - ik) = 0$$
$$\implies r_1 = ik, \; r_2 = -ik \qquad \textit{complex conjugates}$$
$$\implies \lambda = 0, \; \omega = k \qquad \textit{pure imaginary}$$

Linearly independent solutions:

$$y_1 = e^{0 \cdot x} \cos kx = \cos kx\,, \qquad y_2 = e^{0 \cdot x} \sin kx = \sin kx$$

General solution:

$$y = c_1 \cos kx + c_2 \sin kx$$

Homogeneous, **2nd** order, constant coefficients

Example ($y'' - 2y' + y = 0$)

Characteristic equation:

$$r^2 - 2r + 1 = 0 \implies (r-1)^2 = 0 \qquad \textit{real, repeated}$$
$$\implies r = 1$$

Linearly independent solutions:

$$y_1 = e^x\,, \qquad y_2 = x\,e^x$$

General solution:

$$y = (c_1 + c_2 x)e^x$$

A second-order IVP with complex roots

Example ($y'' - 2y' + 5y = 0, \quad y(0) = 1, \quad y'(0) = 3$)

Characteristic equation:

$$r^2 - 2r + 5 = 0 \implies (r-1)^2 = -4 \implies r = 1 \pm 2i.$$

Thus $\lambda = 1, \ \omega = 2.$

Linearly independent solutions:

$$e^x \cos 2x \qquad \text{and} \qquad e^x \sin 2x.$$

General solution:

$$y(x) = e^x(c_1 \cos 2x + c_2 \sin 2x).$$

Now choose c_1, c_2 to satisfy the initial conditions.

A second-order IVP with complex roots

Example ($y'' - 2y' + 5y = 0, \quad y(0) = 1, \quad y'(0) = 3$)

From the preceding slide,

$$y(x) \;=\; e^x(c_1 \cos 2x + c_2 \sin 2x).$$

Differentiate:

$$y'(x) \;=\; e^x\big[(c_1 + 2c_2)\cos 2x + (c_2 - 2c_1)\sin 2x\big].$$

At $x = 0$, the initial conditions give

$$y(0) \;=\; c_1 \;=\; 1,$$
$$y'(0) \;=\; c_1 + 2c_2 \;=\; 3 \implies c_2 \;=\; 1.$$

Solution of the IVP:

$$\boxed{y(x) \;=\; e^x(\cos 2x + \sin 2x)}$$

Higher-order linear ODEs with constant coefficients

Homogeneous, n^{th} order, constant coefficients

Consider

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_1 y' + a_0 y = 0, \qquad (3)$$

where $a_n, \ldots, a_0$ are real constants and $a_n \neq 0$. Try $y = e^{rx}$, as in the second-order case:

$$\begin{aligned} y^{(j)} &= r^j e^{rx}, \\ L[e^{rx}] &= (a_n r^n + a_{n-1} r^{n-1} + \cdots + a_1 r + a_0) e^{rx}. \end{aligned}$$

Since $e^{rx} \neq 0$, we need

$$a_n r^n + a_{n-1} r^{n-1} + \cdots + a_1 r + a_0 = 0.$$

This is the ***characteristic equation***. It is a polynomial equation of degree n.

Roots and solutions

The roots determine a fundamental set of n solutions on $(-\infty, \infty)$.

- Each ***distinct real root*** r_j gives $e^{r_j x}$.
- A real root r with ***multiplicity*** k means that $(z-r)^k$ divides the characteristic polynomial, but $(z-r)^{k+1}$ does not. It gives k independent solutions:

$$e^{rx}, \quad xe^{rx}, \quad \ldots, \quad x^{k-1}e^{rx}.$$

- A pair $\lambda \pm i\omega$, $\omega \neq 0$, each of multiplicity k, gives $2k$ independent solutions:

$$e^{\lambda x}\cos\omega x, \quad xe^{\lambda x}\cos\omega x, \quad \ldots, \quad x^{k-1}e^{\lambda x}\cos\omega x,$$
$$e^{\lambda x}\sin\omega x, \quad xe^{\lambda x}\sin\omega x, \quad \ldots, \quad x^{k-1}e^{\lambda x}\sin\omega x.$$

The general solution is a linear combination of all these functions.

Homogeneous, **4th** order, constant coefficients

Example ($y^{(4)} - 5y'' + 4y = 0$)

Characteristic equation:

$$\begin{aligned} r^4 - 5r^2 + 4 = 0 &\implies (r^2)^2 - 5(r^2) + 4 = 0 \quad \textit{quadratic in } r^2 \\ &\implies (r^2 - 1)(r^2 - 4) = 0 \\ &\implies (r-1)(r+1)(r-2)(r+2) = 0 \\ &\implies r \in \{1, -1, 2, -2\} \end{aligned}$$

Linearly independent solutions:

$$y_1 = e^x, \qquad y_2 = e^{-x}, \qquad y_3 = e^{2x}, \qquad y_4 = e^{-2x}$$

$$W(e^x, e^{-x}, e^{2x}, e^{-2x}) = 72 \neq 0 \qquad \textit{for any } x \in \mathbb{R}$$

General solution: $y = c_1 e^x + c_2 e^{-x} + c_3 e^{2x} + c_4 e^{-2x}, \quad c_i \in \mathbb{R}$

A third-order IVP with a repeated root

Example ($y''' - 3y'' + 4y = 0, \quad y(0) = y'(0) = 0, \quad y''(0) = 3$)

Characteristic equation:

$$\begin{aligned} r^3 - 3r^2 + 4 &= 0 \\ \Longrightarrow \quad (r+1)(r-2)^2 &= 0. \end{aligned}$$

The roots are -1 and 2, with 2 repeated twice.

Linearly independent solutions:

$$e^{-x}, \qquad e^{2x}, \qquad xe^{2x}.$$

General solution:

$$y(x) = c_1 e^{-x} + (c_2 + c_3 x)e^{2x}.$$

There are three constants and three initial conditions.

A third-order IVP with a repeated root

Example ($y''' - 3y'' + 4y = 0, \quad y(0) = y'(0) = 0, \quad y''(0) = 3$)

From the preceding slide,

$$\begin{aligned} y(x) &= c_1 e^{-x} + (c_2 + c_3 x)e^{2x}, \\ y'(x) &= -c_1 e^{-x} + (2c_2 + c_3 + 2c_3 x)e^{2x}, \\ y''(x) &= c_1 e^{-x} + (4c_2 + 4c_3 + 4c_3 x)e^{2x}. \end{aligned}$$

At $x = 0$, the initial conditions give

$$\begin{aligned} c_1 + c_2 &= 0 \implies c_2 = -c_1, \\ -c_1 + 2c_2 + c_3 &= 0 \implies c_3 = 3c_1, \\ c_1 + 4c_2 + 4c_3 &= 3 \implies 9c_1 = 3 \implies c_1 = \frac{1}{3}. \end{aligned}$$

Thus $c_2 = -\frac{1}{3}, \quad c_3 = 1.$

$$\boxed{y(x) = \frac{1}{3}e^{-x} - \frac{1}{3}e^{2x} + xe^{2x}}$$

Practice problems

- Second-order constant-coefficient ODEs [Trench §5.2]:
  - Exercises 1–12: general solutions.
  - Exercises 13–17: initial value problems.
- Higher-order constant-coefficient ODEs [Trench §9.2]:
  - Exercises 1–14: general solutions.
  - Exercises 15, 16, 20–22: initial value problems.